Algebra and Analysis and Topology yay

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1 Root of Unity

1.1 Root of Unity Filter

1.2 Evaluating Sums

2 Inequalities

2.1 Standard Inequalities

2.1.1 Schur's

2.1.2 Cauchy-Schwarz

2.1.3 AM-GM-HM-QM

2.1.4 Weighted AM-GM

2.1.5 Holder's

2.1.6 Power Mean

2.2 Nonstandard Inequalities

Theorem

Holder's Inequality For complex-valued functions f and g and reals p, q satisfying $\frac{1}{p} + \frac{1}{q} = 1$,

$$\|f^p\|^{\frac{1}{p}} \|g^q\|^{\frac{1}{q}} \geq \|fg\|$$

Example

2.3 Smoothing

2.4 Lagrange Multipliers

3 Polynomials + Sequences

3.1 Asymptotic Behavior

Definition

Omega and O notation $O(f(N))$, where $\lim_{N\to\infty} f(n) = \infty$, means that something is bounded by $cf(N)$ as $N \to \infty$ for sufficiently large c. $\Omega(f(n))$ means tha something is on the order of $f(n)$.

Problem

2026 India-Singapore Friendly Contest P5 Suppose that there is a sequence of positive reals $a_0, a_1, \ldots$ such that for all sufficiently large n,

$$\sum_{i=0}^{n} nia_i a_{n-i} \geq 2.01^n \cdot a_n.$$

Show that there exists a positive integer i satisfying $a_i \leq 0.1^{6767}$.

Solution

Assume for contradiction that $a_n \geq c$ for some constant c and all n. Then $\frac{a_i a_{n-i}}{a_n}$ must average to $(1.005)^n$, so a_n is unbounded and $a_n = \Omega((1.005)^n)$. However, this means that a_n is strictly increasing, contradiction.

3.2 Linear Constraints

3.3 Interpolation

4 Misc

4.1 Finite Differences

4.2 Fourier Transformations

Definition

Discrete Fourier Decomposition The decomposition is a systematic breakdown of a function $f(x)$ into sin and cos via

$$f(x) = \sum_{k=0}^{n} a_k \sin(kx) + \sum_{k=0}^{n} b_k \cos(kx)$$

and

$$a_k = \frac{2}{L} \int_0^L f(x) \sin(2\pi nkx) \mathrm{d}x$$

$$b_k = \frac{2}{L} \int_0^L f(x) \cos(2\pi nkx) \mathrm{d}x$$

Definition

Continuous Fourier Decomposition The continuous decomposition of a function $f(x)$ is

$$[x^d]f(x) = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{-id\theta} f(e^{i\theta})\mathrm{d}\theta$$

Problem

2025 CTST 15 Let X be a finite set of real numbers, d be a real number, and $\lambda_1, \lambda_2, \cdots, \lambda_{2025}$ be 2025 non-zero real numbers. Define

$$A = \left\{(x_1, x_2, \cdots, x_{2025}) : x_1, x_2, \cdots, x_{2025} \in X and \sum_{i=1}^{2025} \lambda_i x_i = d\right\},$$

$$B = \left\{(x_1, x_2, \cdots, x_{2024}) : x_1, x_2, \cdots, x_{2024} \in X and \sum_{i=1}^{2024} (-1)^i x_i = 0\right\},$$

$$C = \left\{(x_1, x_2, \cdots, x_{2026}) : x_1, x_2, \cdots, x_{2026} \in X and \sum_{i=1}^{2026} (-1)^i x_i = 0\right\}.$$

Show that $|A|^2 \leq |B| \cdot |C|$.

Solution

Define $f(t) = \sum_{x\in X} t^x$ to be a generating function. Then

$$|A| = \lim_{T\to\infty} \frac{1}{T}\int_0^T e^{-id\theta} \prod_{j=1}^{2025} f(e^{i\lambda_j\theta})\mathrm{d}\theta$$

$$|B| = \lim_{T\to\infty} \frac{1}{T}\int_0^T f(e^{-i\theta})^{1012} f(e^{i\theta})^{1012}\mathrm{d}\theta = \lim_{T\to\infty} \frac{1}{T}\int_0^T |f(e^{i\theta})^{2024}\mathrm{d}\theta|$$

$$|C| = \lim_{T\to\infty} \frac{1}{T}\int_0^T f(e^{-i\theta})^{1013} f(e^{i\theta})^{1013}\mathrm{d}\theta = \lim_{T\to\infty} \frac{1}{T}\int_0^T |f(e^{i\theta})^{2026}\mathrm{d}\theta$$

so

$$|B||C| \geq \left(\lim_{T\to\infty} \frac{1}{T}\int_0^T |f(e^{i\theta})^{2025}|\mathrm{d}\theta\right)^2$$

by Cauchy-Schwarz. To deal with A, write

$$|A| = \lim_{T\to\infty} \frac{1}{T}\int_0^T e^{-id\theta} \prod_{j=1}^{2025} f(e^{i\lambda_j\theta})\mathrm{d}\theta$$

Solution

$$\leq \left(\lim_{T\to\infty} \frac{1}{T} \prod_{j=1}^{2025} \int_0^T \left(e^{-id\theta} f(e^{i\lambda_j \theta}) \right)^{2025} \mathrm{d}\theta \right)^{\frac{1}{2025}} \leq \lim_{T\to\infty} \frac{1}{T} \int_0^T |f(e^{i\theta})^{2025}| \mathrm{d}\theta$$

as desired

5 Analysis

Theorem

Gauss-Lucas Given a polnyomial $f(z) \in \mathbb{C}[x]$ with roots $z_1, z_2, \ldots, z_n$, its derivative $f'(z)$ with roots $w_1, w_2, \ldots, w_{n-1}$ lie within the convex hull of $z_1, \ldots, z_n$

Assume that $f(x)$ shares no roots with $'f(x)$ by dividing by roots. Consider

$$\frac{d}{dz} \ln f(z) = \frac{f'(z)}{f(z)} = \sum_{i=1}^{n} \frac{1}{z - z_i}.$$

Set $z \in \{w_1, \ldots, w_{n-1}\}$ to get

$$\sum_{i=1}^{n} \frac{1}{w_j - z_i} = 0 \iff w_j \sum_{i=1}^{n} \frac{1}{|w_j - z_i|^2} = \sum_{i=1}^{n} \frac{z_i}{|w_j - z_i|^2}$$

$$\iff w_j = \frac{\sum_{i=1}^{n} \frac{z_i}{|w_j - z_i|^2}}{\sum_{i=1}^{n} \frac{1}{|w_j - z_i|^2}}$$

meaning that w_j is a convex sum of the vectors $z_1, \ldots, z_n$ as needed.

Definition

Point Derivative The point derivative of a complex-valued function $f(z)$ is

$$D_\zeta f(z) = (\zeta - z) f'(z) + n f(z)$$

where $\deg(f) = n$.

Proposition

Gauss-Lucas for Point Derivative Let the roots of $D_\zeta f(z)$ be $w_1, \ldots, w_{n-1}$ and $f(z)$ be $z_1, \ldots, z_n$. Then $\max(\|w_1\|, \ldots, \|w_{n-1}\|) \leq \max(\|z_1\|, \ldots, \|z_n\|)$.

Let w be a root of $D_\zeta f(z)$, so

$$\frac{f'(z)}{f(z)} = \frac{d}{dz} \ln f(z) = \sum_{i=1}^{n} \frac{1}{z_i - \zeta} = \frac{n}{z - \zeta}$$

However $z \to \frac{1}{z-\zeta}$ is a mobius transformation sending a disc to another disc, and the sum of many discs is convex. We therefore have equality when z_i are identical.

Problem

2026 CTST 3 Let integers $n > k > 1$, and let $z_1, z_2, \ldots, z_n$ be complex numbers with modulus at most 1. Prove that

$$\left| nk - \sum_{1 \leq i_1 < i_2 < \cdots < i_k \leq n} z_{i_1} z_{i_2} \cdots z_{i_k} \right| \leq n - 1k - 1 \left| n - \sum_{i=1}^{n} z_i \right|,$$

and find the condition for equality.

Solution

Claim: It suffices to consider the case $n = k$ *Proof.* Let $P'(z)$ be the derivative of $P(z)$, and suppose that the claim holds for $(n-1, k)$. We will show it holds for (n, k). Let $P(z) = z^n + a_{n-1}z^{n-1} + \ldots + a_0 = (z - z_1)(z - z_2)\ldots$ so that $P'(z) = nz^{n-1} + \ldots + a_1 = n(z - \omega_1)(z - \omega_2)\ldots(z - \omega_{n-1})$. Then

$$\left| \binom{n-1}{k-1} - \frac{a_{n-k}}{n} \right| \leq \binom{n-2}{k-2} \left| n - 1 - \frac{a_{n-1}}{n} \right|$$

$$\left| \binom{n}{k} - \sum_{1 \leq i_1 < \ldots < i_k < n} z_{i_1} \ldots z_{i_k} \right| \leq \frac{k-1}{k} \binom{n-1}{k-1} \left| n - \sum_{i=1}^{n} z_i \right|$$

as desired. Note that there is an injective mapping between every polynomial with degree n and its derivative, so this suffices because the roots $\omega_1, \omega_2, \ldots, \omega_{n-1}$ must be contained within the hull of $z_1, z_2, \ldots, z_n$ due to Gauss-Lucas.

Solution

Claim: $|1-\prod_{i=1}^{n} z_i| \leq |n-\sum_{i=1}^{n} z_i|$ and equality occurs when $z_i = 1$ for all $1 \leq i \leq n$ *Proof.* Consider the point derivative function $D_\zeta f(z) = (\zeta - z)f'(z) + nf(z)$ where n is the degree of z. Consider the polynomial $P(z) = (z-1)^n + a_{n-1}z^{n-1} + \ldots + a_0$ and note that $|a_0| = |1 - \prod z_i|$ and $|a_{n-1}| = |n - \sum z_i|$. Suppose for the sake of contradiction that $|a_{n-1}| < |a_0| \leq 1$. If the roots of z have modulus at most 1, then the point derivative must map $(z_1, z_2, \ldots, z_n) \to (\omega_1, \ldots, \omega_{n-1})$ such that $\max(\omega_i) \leq 1$. We apply the derivative once to remove the $(z-1)^n$ term and are left with $D_1 f(z) = a_{n-1}z^{n-1} + ((n-1)a_{n-1} + 2a_{n-2})z^{n-2} + \ldots + (a_k(n-k) + a_{k+1}(k+1)) + \ldots + (a_1 + na_0)$ where all its roots lie within the unit disc. Then we will show that $|a_k| \geq \binom{n-1}{k}|a_0|$, meaning that $|a_{n-1}| \geq |a_0|$, giving the result. We know that $|a_1| \geq |a_0 - na_0| = |a_0|(n-1)$. Then if $|a_k| \geq \binom{n-1}{k}|a_0|$, then $|(n-k)a_k + (k+1)a_{k+1}| \leq 1a_{k+1} \geq \binom{n-1}{k}(n-k)|a_0| - |a_0| \geq \binom{n-1}{k+1}|a_0|$. Therefore, $a_{n-1} \geq |a_0|$, a contradiction with equality when $a_0 = 0$. From here, it's easy to check that $(z_1, z_2, \ldots, z_n) = (\underbrace{1, 1, \ldots}_{n})$ is the only equality case.

Problem

2018 Putnam B2 Let n be a positive integer, and let $f_n(z) = n + (n-1)z + (n-2)z^2 + \ldots + z^{n-1}$. Prove that f_n has no roots in the closed unit disk $\{z \in \mathbb{C} : |z| \leq 1\}$.

Solution

It suffices to show $z^{n-1}f_n(\frac{1}{z})$ has no roots $z > 1$. Integrate to get roots of unity, so by Gauss-Lucas, all roots must be contained within the unit disc, contradiction.

6 Algebraic Geometry

Theorem

Riemann-Roch

Theorem

Hasse-Weil Given a curve polynomial curve $C : y^2 = f(x)$ evaluated over a finite field $\mathbb{F}_{q^k}$, we necessarily have

$$|q^k + 1 - \#C(\mathbb{F}_{q^k})| \leq 2g\sqrt{qa^k}$$

where g is the genus of the curve.

Proof